

Alessandro Giacchetto

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Employment

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|--------------|---|
| 2026–present | Rita Levi-Montalcini Fellow , University of Padua |
| 2023–2026 | ETH Fellow & Hermann-Weyl-Instructor , ETH Zurich
Mentor: Rahul Pandharipande |
| 2021–2023 | Postdoctoral Fellow , IPhT
Mentor: Bertrand Eynard |

Education

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| 2017–2021 | PhD in Mathematics , MPIM
<i>Geometric and topological recursion and invariants of the moduli space of curves</i>
Supervisor: Gaëtan Borot |
| 2015–2017 | MSc in Mathematics , University of Trieste & SISSA
<i>The J-equation on Kähler manifolds and blowups</i>
Supervisor: Jacopo Stoppa |
| 2012–2015 | BSc in Mathematics , University of Trieste
<i>Skein relations and polynomial invariants of knots and links</i>
Supervisor: Bruno Zimmermann |

Publications and preprints

Euler characteristics of strata of k -differentials, with M. Costantini, M. Möller, A. Sauvaget, J. Schmitt ·

Preprint · 2026 · [arXiv](#)

Membranes and maps, with R. Pandharipande, Y. Schuler ·

Preprint · 2026 · [arXiv](#)

A bijective topological recursion for maps, with G. Borot, L. Merkens ·

Preprint · 2026 · [arXiv](#)

Refined lattice point counting on the moduli space of Klein surfaces, with N. K. Chidambaram, E. Garcia-Failde, K. Osuga ·

Preprint · 2026 [arXiv](#)

Worksheet duals to one-matrix models, with R. Gopakumar, E. A. Mazenc ·

Preprint · 2026 [arXiv](#) [Notebook](#)

Matrix correlators as discrete volumes of moduli space I: recursion relations, the BMN-limit and DSSYK, with P.

Maity, E. A. Mazenc ·

Preprint · 2025 [arXiv](#)

Theta classes: generalized topological recursion, integrability and $\mathcal{W}$ -constraints, with V. Bouchard, N. K. Chidambaram, S.

Shadrin ·

Preprint · 2026 [arXiv](#)

Les Houches lecture notes on moduli spaces of Riemann surfaces, with D. Lewański ·

Les Houches Summer School Lecture Notes - 2024-08: Quantum Geometry · *SciPost Phys. Lect. Notes* · 2025 [arXiv](#) [Open access](#)

The factorial growth of topological recursions, with G. Borot, B. Eynard ·

Lett. Math. Phys. 115.62 · 2025 [arXiv](#) [DOI](#)

Can transformers do enumerative geometry?, with B. Hashemi, R. G. Corominas ·

13th Int. Conf. Learn. Represent. · 2025 [arXiv](#) [Open access](#) [Code](#)

Symmetries of F-cohomological field theories and F-topological recursion, with G. Borot, G. Umer ·

Commun. Math. Phys. 406.248 · 2025 [arXiv](#) [DOI](#)

Length spectrum of large genus random metric maps, with S. Barazer, M. Liu ·

Forum Math. Sigma 13 · 2025 [arXiv](#) [DOI](#)

Resurgent large genus asymptotics of intersection numbers, with B. Eynard, E. Garcia-Failde, P. Gregori, D. Lewański ·

Preprint · 2026 [arXiv](#) [DOI](#)

The spin Gromov–Witten/Hurwitz correspondence for $\mathbb{P}^1$, with R. Kramer, D. Lewański, A. Sauvaget ·

J. Eur. Math. Soc. · 2025 [arXiv](#) [DOI](#)

Relations on $\overline{\mathcal{M}}_{g,n}$ and the negative r -spin Witten conjecture, with N. K. Chidambaram, E. Garcia-Failde ·

Invent. Math. 241 · 2025 [arXiv](#) [DOI](#)

Shifted Witten classes and topological recursion, with S. Charbonnier, N. K. Chidambaram, E. Garcia-Failde ·

Trans. Amer. Math. Soc. 377.2 · 2024 [arXiv](#) [DOI](#)

An intersection-theoretic proof of the Harer–Zagier formula, with D. Lewański, P. Norbury ·

Algebraic Geom. 10.2 · 2023 [arXiv](#) [DOI](#)

Around the combinatorial unit ball of measured foliations on bordered surfaces, with G. Borot, S. Charbonnier, V. Delecroix, C. Wheeler ·

Int. Math. Res. Not. 2023.17 · 2023 [arXiv](#) [DOI](#)

A new spin on Hurwitz theory and ELSV via theta characteristics, with R. Kramer, D. Lewański ·

Selecta Math. 31.90 · 2025 [arXiv](#)

On the Kontsevich geometry of the combinatorial Teichmüller space, with J. E. Andersen, G. Borot, S. Charbonnier, D.

Lewnański, C. Wheeler ·

Accepted in *C.I.M.E. Lecture Notes* · 2026  [arXiv](#)

Masur–Veech volumes and intersection theory, the principal strata of quadratic differentials, with G. Borot, D.

Lewnański ·

Appendix in *Duke Math. J.* 172.9 · 2023  [arXiv](#)  [DOI](#)

Topological recursion for Masur–Veech volumes, with J. E. Andersen, G. Borot, S. Charbonnier, V. Delecroix, D. Lewnański, C.

Wheeler ·

J. London Math. Soc. 107.1 · 2023  [arXiv](#)  [DOI](#)

Teaching

Topological recursion and intersection theory, PhD mini-course, UIUC, Fall 2026

Introduction to topological recursion, PhD mini-course, ICTS Bengaluru, Summer 2026

Moduli spaces of Riemann surfaces, PhD mini-course, KMI Nagoya, Winter 2026

[Toric Geometry](#), lecturer, ETHZ, Spring 2025

Moduli spaces of Riemann surfaces, PhD mini-course, Les Houches, Summer 2024

[Riemann Surfaces](#), lecturer, ETHZ, Spring 2024

Hurwitz numbers and Gromov–Witten theory, PhD mini-course, Hiddensee, Summer 2022

Topology, teaching assistant, UniTS, Spring 2017

Linear algebra, teaching assistant, UniTS, Fall 2016

Organization of events

Workshop [Refinement in Enumerative Geometry and Physics](#), Bangalore (22 Jun–3 Jul, 2026)

School [Quantum Geometry](#), Les Houches (29 Jul–23 Aug, 2024)

School [Topological recursion and integrability](#), Trieste (11–16 Sep, 2023)

Conference [Moduli spaces: theory and coding](#), Les Diablerets (27 Feb–3 Mar, 2023)

School + workshop [TR Salento 2022](#), Otranto (5–16 Sep, 2022)

Conference [Recent advances on moduli spaces of curves](#), Leysin (18–24 Mar, 2022)

Reading group [Donaldson–Thomas invariants](#), MPIM (Spring 2020)

Reading group [Integer points in polyhedra](#), MPIM (Fall 2019)

Reading group [Geometric recursion](#), MPIM (Fall 2018)

Honours

2026–2030	Rita Levi-Montalcini Fellowship
2025	SwissMAP Junior Researcher Prize
2023–2025	ETH Fellowship
2018	Marco Reni prize
2015–2017	SISSA MSc Fellowship
2012/13	Best freshman, University of Trieste
2012–2015	Luciano Fonda BSc Fellowship